Haoming (Neil) Li

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Work authorization: CPT eligible for internships

Research Interests

My current focus is **personal AI** and **physical AI**, and the **agentic memory** that both rely on. Before this, I worked on **computer-use agents (CUAs)**, including error recovery in long-horizon tasks and joint reinforcement learning of multiple CUAs. Earlier, I worked on LLM planning and scientific summarization.

Education

The University of Texas at Dallas, Ph.D. in Computer Science, GPA 3.83/4.00 — Aug 2025 – Present
Emory University, M.S. in Computer Science, GPA 3.97/4.00 — Aug 2022 – May 2024
Renmin University of China, B.S. in Data Science and B.Econ. in Finance — Sep 2018 – Jun 2022

Publications

†equal contribution

[1] **Haoming Li**†, Defu Cao†, Xu Hu†, Yingtao Luo, Yiyang Wang, Xianghe Xu, Jiashu He, Wanting Jiang, Yifan Zhang, Yunhan Zhao, Siyang Wu, Chen Yang, Haixin Wang, Zheyuan Deng, Taoran Li, et al. *PersonaCUABench: Benchmarking Computer-Use Agents Serving Diverse Personas.* Under review at **ACL ARR October 2026**. [arXiv]

[2] **Haoming Li**, Dingyi Kang, Xu Hu, Jinxin Yang, Zhu Xu, Chenru Wang, Yi Ge, Miao Yin, Guanpeng Li, Feng Chen, Bingzhe Li. *SAFE-Mem: Secure Agentic Memory with Fine-Grained Authority and Governance-Aware Retrieval.* Under review at **ACL ARR October 2026**. [arXiv]

[3] Xu Hu†, **Haoming Li**†, Yuxiang Wang, Jinxin Yang, Kangshuo Li, Licheng Wang, Dingyi Kang, Jiayu Song, Tianhao Wang, Zhanhua Hu, Xu Yang, Qiannan Li, Bingzhe Li, Feng Chen. *CUA-Recovery: Benchmarking Error Recovery in Long-Horizon Computer-Use Agents.* Under review at **ICLR 2027**. [arXiv]

[4] Siyang Wu, Weizhi Zhang, Chen Yang, Esther Xu Fei, Jorin Overwiening, Xu Pan, Jing Tang, Yixuan He, Kelvin C. Mo, Peiwen Li, Defu Cao, Xu Hu, Haixin Wang, Taoran Li, **Haoming Li**, et al. *PersonaScienceBench: Evaluating Scientific Agents with Diverse Simulated Researchers.* Under review at **ACL ARR October 2026**.

[5] Xu Hu, **Haoming Li**, Sabbir Ahmed, Md Nahiyan Uddin, Qiannan Li, Jessica Ouyang, Latifur Khan, Feng Chen, Bingzhe Li. *Toward Trustworthy Computer-Use Agents: Risk Propagation, Evaluation Gaps, and Human Governance.* Under review at **ACM Computing Surveys**.

[6] Weili Shi, **Haoming Li**, Sheng Li. *Enhancing Dynamic Text-Attributed Graph Modeling via Instruction Tuning.* **Findings of AACL-IJCNLP 2026** (to appear).

[7] **Haoming Li**, Jessica Ouyang. *GRASP: Graph-Reasoning Aided Survey Planning for High-Fidelity Related Work Generation.* **Findings of ACL 2026**. [paper]

[8] **Haoming Li**, Zhaoliang Chen, Jonathan Zhang, Fei Liu. *PLANET: A Collection of Benchmarks for Evaluating LLMs' Planning Capabilities.* arXiv:2504.14773, 2025.

[9] **Haoming Li**, Zhaoliang Chen, Songyuan Liu, Yiming Lu, Fei Liu. *Systematic Analysis of LLM Contributions to Planning: Solver, Verifier, Heuristic.* arXiv:2412.09666, 2024.

[10] **Haoming Li**, Zhaoliang Chen, Jonathan Zhang, Fei Liu. *LASP: Surveying the State-of-the-Art in Large Language Model-Assisted AI Planning.* arXiv:2409.01806, 2024.

[11] Yikun Wang, Rui Zheng, **Haoming Li**, Qi Zhang, Tao Gui, Fei Liu. *RESCUE: Ranking LLM Responses with Partial Ordering to Improve Response Generation.* **ACL 2024 Student Research Workshop**. [paper]

Research Experience

MatrAIx, Research Collaborator (persona agents) — Sep 2026 – Present

- **PersonaCUABench** (co-first author) [1]: selected and designed 44 personas in four groups across 11 domains, and evaluated nine CUA backbones (3,330 runs). Persona-phrased requests cut mean task completion from 0.689 to 0.441 (36% relative), and even fully completed persona requests receive the top satisfaction rating in only 62.7% of cases, against 84.1% for the original requests.
- Contributed to **PersonaScienceBench** [4] and the **PersonalAgentEval** leaderboard for personal agents across diverse users, preferences, and domains.

The University of Texas at Dallas, Research Assistant (advisors: Feng Chen, Bingzhe Li) — Jun 2025 – Present

- **SAFE-Mem** (first author) [2]: designed an agentic memory that accounts for both the stored content and its information source, making memory both secure and accurate. Across four LLM backbones on MPBench and LoCoMo, it has the lowest attack success among defenses that keep legitimate memory (at least 77% below the strongest of them) and the highest answer accuracy of all evaluated defenses (up to 15% higher).
- **CUA-Recovery** (co-first author) [3]: designed *ReRail*, the task and trajectory synthesis framework behind the benchmark (2,250 human-verified erroneous states for testing, 8.6K trajectories for training).

- Analyzed six agents at six takeover depths (best error-awareness rate: 44.0%) and fine-tuned Qwen3.5-35B-A3B on the training split with SFT, raising its error-awareness rate from 19.9% to 35.2% on held-out states.
- **Trustworthy CUA survey** (second author) [5]: reviewed and organized the literature on CUA security, privacy, reliability, controllability, and accountability.
- **GRASP** (first author, advised by Jessica Ouyang) [7]: designed, implemented, and evaluated a graph-reasoning planner for related-work generation.

University of Virginia, Reasoning & Knowledge Discovery Lab, Research Assistant (advisor: Sheng Li) Jun 2024 – May 2025

- Instruction-tuned LLMs to encode and reason over dynamic text-attributed graphs [6].

Emory University, AI/LLM Agents Research Group, Research Assistant (advisor: Fei Liu) Mar 2023 – May 2025

- **LLMs for planning** (first author) [8, 9, 10]: led an empirical study of LLMs as solver, verifier, and heuristic function in planning, a collection of planning benchmarks for LLMs, and a survey of LLM-assisted planning.
- **RESCUE** [11]: trained a response ranker from partial orderings over candidate LLM outputs to improve response generation.

Service and Skills

Reviewer: NeurIPS 2026, IEEE BIBM 2026, International Journal of Data Science and Analytics (IJDSA).
Languages: English (fluent), Chinese (native).
Programming: Python, PyTorch, C/C++, SQL, Bash. **Tools:** Hugging Face Transformers, vLLM, Docker, Spark.